Numerical methods for PDE

Many physical phenomena can be modelled mathematically by D.Es. When there are two or more independent variables & one dependent variable, the D.E is a PDE. For some PDE getting analytical solution is difficult. There are various methods to get approximate solns to the PDEs. Here we are discussing the finite difference approximation method to solve PDE.

Classification of 2nd order Equation

The general 2nd order linear PDE in 2 independent variables x, y is of the form

$$A\frac{\partial^2 u}{\partial x^2} + B\frac{\partial^2 u}{\partial x\,\partial y} + C\frac{\partial^2 u}{\partial y^2} + D\frac{\partial u}{\partial x} + E\frac{\partial u}{\partial y} + Fu = 0$$

Where A, B, C, D, E & F are general functions of x & y. This equation is said to be

(i) Parabolic if $B^2 - 4AC = 0$

(ii) Elliptic if $B^2 - 4AC < 0$

(iii) Hyperbolic if $B^2 - 4AC > 0$

① $3u_{xx} + 10u_{xy} + 3u_{yy} = 0$. Hyperbolic

② $x^2u_{xx} - 2xyu_{xy} + y^2u_{yy} = e^x$. Parabolic

③ $u_{xx} + x^2u_{yy} = 0$. elliptic

④ $y^2u_{xx} - 2xyu_{xy} + x^2u_{yy} \rightarrow$ parabolic

⑤ $u_{xx} + xu_{yy} = 0$

⑥ $yu_{xx} + (x+y)u_{xy} + xu_{yy} = 0$

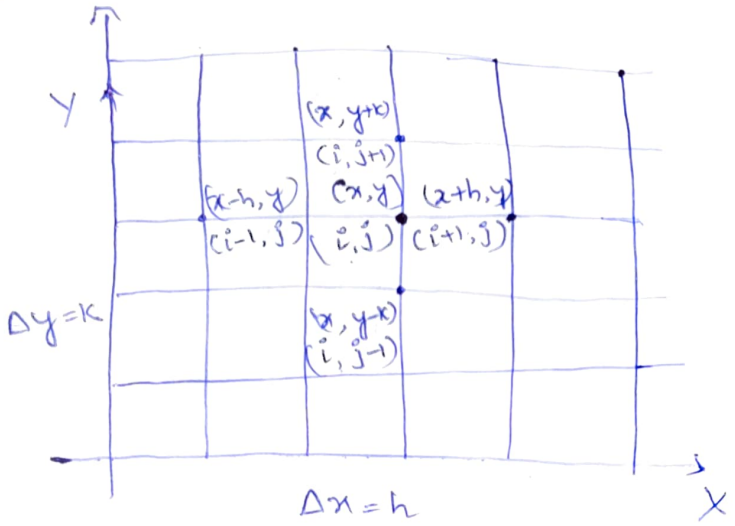
$x+y^2 - 4xy$

$= (x-y)^2 > 0$

Hyperbolic

Solution of PDEs - Finite difference approximations to Partial derivatives

Consider a rectangular region R in the x-y plane. Divide this region into a rectangular network of sides $\Delta x = h$ & $\Delta y = k$ as shown in the fig. The points of intersection are called mesh points, nodal points or grid points



Then the finite difference approximations for the partial derivatives w.r.t x are

$$\frac{\partial u}{\partial x} = \frac{u(x+h,y) - u(x,y)}{h} + O(h)$$ forward difference approximation (FDA)

$$= \frac{u(x,y) - u(x-h,y)}{h} + O(h)$$ backward —"— (BDA)

$$= \frac{u(x+h,y) - u(x-h,y)}{2h} + O(h^2)$$ central —"— (CDA)

& $$\frac{\partial^2 u}{\partial x^2} = \frac{u(x-h,y) - 2u(x,y) + u(x+h,y)}{h^2} + O(h^2)$$

Writing $u(x,y) = u(ih, jk)$ as simply $u_{i,j}$ the

above approximations become

$$u_x = \frac{u_{i+1,j} - u_{ij}}{h} + O(h) \quad \text{FDA}$$

$$= \frac{u_{i,j} - u_{i-1,j}}{h} + O(h) \quad \text{BDA}.$$

$$= \frac{u_{i+1,j} - u_{i-1,j}}{2h} + O(h^2) \quad \text{CDA}$$

$$\& \; u_{xx} = \frac{u_{i-1,j} - 2u_{i,j} + u_{i+1,j}}{h^2} + O(h^2)$$

Illy the approximations for the derivative w.r.t y

$$u_y = \frac{u_{i,j+1} - u_{i,j}}{k} + O(k)$$

$$= \frac{u_{i,j} - u_{i,j-1}}{k} + O(k)$$

$$= \frac{u_{i,j+1} - u_{i,j-1}}{2k} + O(k^2)$$

$$\& \; u_{yy} = \frac{u_{i,j-1} - 2u_{i,j} + u_{i,j+1}}{k^2} + O(k^2)$$

Replacing the derivatives in any PDE by their corresponding difference approximations given in above formulae

Numerical soln of one dimensional heat equation (Parabolic eqn)

The standard equation is of the form

$$\frac{\partial u}{\partial t} = c^2 \frac{\partial^2 u}{\partial x^2} \quad \text{where } c^2 = \frac{k}{s\rho} \text{ is the} \longrightarrow (1)$$

diffusivity of the substance

Consider a rectangular mesh in x-t plane with spacing h along x direction & k along time t direction. Denoting mesh point $(x,t) = (i,j)$ simply. we have

$$\frac{\partial u}{\partial t} = \frac{u_{i,j+1} - u_{i,j}}{k} \quad \& \quad \frac{\partial^2 u}{\partial x^2} = \frac{u_{i+1,j} - 2u_{i,j} + u_{i+1,j}}{h^2}$$

then (1) reduces to

$$u_{i,j+1} = \alpha u_{i-1,j} + (1-2\alpha) u_{i,j} + \alpha u_{i+1,j} \longrightarrow (2)$$

where $\alpha = \frac{kc^2}{h^2}$

(2) is called Schmidt's explicit 2 level formula. It is valid for $0 < \alpha \leq \frac{1}{2}$

In particular if $\alpha = \frac{1}{2}$ (2) reduces to

$$u_{i,j+1} = \frac{1}{2}\left[u_{i-1,j} + u_{i+1,j}\right]$$ This is called Bendre-Schmidt method. For $\alpha > \frac{1}{2}$ the method is divergent & unstable.

$u_{i,j+1}$

$\alpha(u_{i-1,j})$ $(1-2\alpha)u_{i,j}$ $\alpha(u_{i+1,j})$

$\alpha = 1/2$

$u_{i,j+1}$

$\frac{1}{2}(u_{i+1,j})$ $\frac{1}{2}(u_{i+1,j})$

Working rule

(1) Form the basic informative table based on given step sizes h & k.

(2) If k is not given, find $k = \frac{h^2}{2c^2}$ (If it is Bendre - Schmidt method)

(3) Write the values of u_{ij} with reference to the conditions $u(0,t) = 0$ & $u(l,t) = 0$. (i.e x varies from 0 to l, values along I & last column are zero)

(4) The values along the first row are obtained by direct computation using $u(x,0) = f(x)$.

(5) use $u_{i,j+1} = \frac{1}{2}[u_{i-1,j} + u_{i+1,j}]$ if $\alpha = \frac{1}{2}$

(Or) $u_{i,j+1} = \alpha[u_{i-1,j} + u_{i+1,j}] + (1-2\alpha)u_{ij}$ for $\alpha < \frac{1}{2}$ where $\alpha = \frac{kc^2}{h^2}$

(1) Find the soln of parabolic eqn $u_{xx} = 2u_t$. where $u(0,t) = u(4,t) = 0$. & $u(x,0) = x(4-x)$ taking $h = 1$ find the values upto $t = 5$.

Soln Compare $u_{xx} = 2u_t$ with $u_t = c^2 u_{xx}$.

$c^2 = \frac{1}{2}$, $h = 1$, $k = 1$,

$\alpha = \frac{kc^2}{h^2} = \frac{1(\frac{1}{2})}{1} = \frac{1}{2}$

Use Bendre - Schmidt method

Given BC's $u(0,t) = u(4,t) = 0$ i.e $0 \le x \le 4$

means when $x = 0$ & $x = 4$ the values of u are zero i.e I & last column are zero

Initial Condition $u(x,0) = x(4-x)$.

When $x = 1$, $u(1,0) = 1(4-1) = 3$

$x = 2$, $u(2,0) = 2(4-2) = 4$

$x = 3$, $u(3,0) = 3(4-3) = 3$

t \ x	0	1	2	3	4
0	$u_{00}=0$	$u_{10}=3$	$u_{20}=4$	$u_{30}=3$	$u_{40}=0$
1	$u_{01}=0$	$u_{11}=2$	$u_{21}=3$	$u_{31}=2$	$u_{41}=0$
2	$u_{02}=0$	$u_{12}=1.5$	$u_{22}=2$	$u_{32}=1.5$	$u_{42}=0$
3	$u_{03}=0$	$u_{13}=1$	$u_{23}=1.5$	$u_{33}=1$	$u_{43}=0$
4	$u_{04}=0$	$u_{14}=0.75$	$u_{2,4}=1$	$u_{3,4}=0.75$	$u_{44}=0$
5	$u_{0,5}=0$	$u_{1,5}=0.5$	$u_{2,5}=0.75$	$u_{3,5}=0.5$	$u_{45}=0$

2nd, 3rd, 4th & 5th & 6th rows are obtained by $u_{i,j+1} = \frac{1}{2}\left[u_{i-1,j} + u_{i+1,j}\right]$

i.e For II row

$u_{11} = \frac{1}{2}\left[u_{00} + u_{20}\right] = 2$

$u_{21} = \frac{1}{2}\left[u_{10} + u_{30}\right] = 3$ $u_{31} = \frac{1}{2}\left[u_{20} + u_{40}\right] = 2$

IIIly other rows [$u_{11} \rightarrow$ go up [add left & right] $\div 2$]

H.W

(2) Find the Values of $u(x,t)$ satisfying the parabolic eqn $\frac{\partial u}{\partial t} = 4\frac{\partial^2 u}{\partial x^2}$ & the boundary condition $u(0,t) = 0 = u(8,t)$ & $u(x,0) = 4x - \frac{1}{2}x^2$ at the points $x = i : i = 0, 1, 2, \ldots 8$ & $t = \frac{1}{8} j :$

$j = 0, 1, 2, 3, 4, 5$

t \ x	0	1	2	3	4	5	6	7	8
0 | 0	u_{00}	u_{10}	u_{20}	u_{30}	u_{40}	u_{50}	u_{60}	u_{70}	u_{80}
1/8 | 1	u_{01}	u_{11}	u_{21} =	u_{31} =	u_{41} =	u_{51} =	u_{61}	u_{71}	u_{81}
2/8 | 2	u_{02}	u_{12}	u_{22}	u_{32}	u_{42}	u_{52}	u_{62}	u_{72}	u_{82}
3/8 | 3	u_{03}	u_{13}	u_{23}	u_{33}	u_{43}	u_{53}	u_{63}	u_{73}	u_{83}
4/8 | 4	u_{04}	u_{14}	u_{24}	u_{34}	u_{44}	u_{54}	u_{64}	u_{74}	u_{84}
5/8 | 5	u_{05}	u_{15}	u_{25}	u_{35}	u_{45}	u_{55}	u_{65}	u_{75}	u_{85}

(3) Solve $\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2}$ subject to conditions

$u(x,0) = \sin \pi x \quad 0 \le x \le 1, \quad u(0,t) = u(1,t) = 0$.

Carry out computations for 2 levels taking

$h = \frac{1}{3}, \quad k = \frac{1}{36}$.

Soln Compare $\frac{\partial u}{\partial t} = c^2 \frac{\partial^2 u}{\partial x^2}$ with $\frac{\partial u}{\partial t} = c^2 \frac{\partial^2 u}{\partial x^2}$

$c^2 = 1, \quad h = \frac{1}{3} \quad k = \frac{1}{36}$

$$\alpha = \frac{kc^2}{h^2} = \frac{\frac{1}{36}(1)}{\frac{1}{9}} = \frac{1}{36} \times 9 = \frac{1}{4} < \frac{1}{2}$$

Boundary conditions : $u(0,t) = u(1,t) = 0$.

Initial conditions : $u(x,0) = \sin \pi x$

t \ x	0	$\frac{1}{3}$	$\frac{2}{3}$	$\frac{3}{3} = 1$
	0	1	2	3
0	0	$u_{00} = 0$	$u_{10} = 0.866$	$u_{20} = 0.866$	$u_{30} = 0$
$\frac{1}{36}$	1	$u_{01} = 0$	$u_{11} = 0.6495$	$u_{21} = 0.6495$	$u_{31} = 0$
$\frac{2}{36}$	2	$u_{02} = 0$	$u_{12} = 0.4871$	$u_{22} = 0.4871$	$u_{32} = 0$

Carry out 2 levels means $0 \to \frac{1}{36} \to \frac{2}{36}$ (1, 2)

I row :

Since $u(x,0) = \sin \pi x$.

$x = \frac{1}{3}, \quad u(\frac{1}{3}, 0) = \sin \frac{\pi}{3} = \frac{\sqrt{3}}{2} = 0.866$

$x = \frac{2}{3} \quad u(\frac{2}{3}, 0) = \sin \frac{2\pi}{3} = \frac{\sqrt{3}}{2} = 0.866$

Ex

II row

$$u_{i,j+1} = \alpha\left[u_{i-1,j} + u_{i+1,j}\right] + (1-2\alpha)u_{ij}$$

$$1-2\alpha = 1-2\left(\frac{1}{4}\right) = 1-\frac{1}{2} = \frac{1}{2}$$

$$= \frac{1}{4}\left[u_{i-1,j} + u_{i+1,j}\right] + \frac{1}{2}u_{i,j}$$

$$\Rightarrow \frac{1}{4}\left[u_{i-1,j} + u_{i+1,j} + 2u_{i,j}\right]$$

$$\therefore u_{\frac{1}{3},\frac{1}{36}} = \frac{1}{4}\left[u_{00} + u_{\frac{2}{3},0} + 2u_{\frac{1}{3},0}\right]$$

$$= \frac{1}{4}\left[0 + 0.866 + 2(0.866)\right]$$

$$= 0.6495$$

$$u_{\frac{2}{3},\frac{1}{36}} = \frac{1}{4}\left[u_{\frac{1}{3},0} + u_{1,0} + 2u_{\frac{2}{3},0}\right]$$

$$= \frac{1}{4}\left[0.866 + 0 + 2(0.866)\right]$$

$$= 0.6495$$

III row.

$$u_{\frac{1}{3},\frac{2}{36}} = \frac{1}{4}\left[u_{0,\frac{1}{36}} + u_{\frac{2}{3},\frac{1}{36}} + 2(0.6495)\right]$$

$$= \frac{1}{4}\left[0 + 0.6495 + 2(0.6495)\right]$$

$$= 0.4871$$

$$u_{\frac{2}{3},\frac{2}{36}} = \frac{1}{4}\left[u_{\frac{1}{3},\frac{1}{36}} + u_{1,\frac{1}{36}} + 2u_{\frac{2}{3},\frac{1}{36}}\right]$$

$$= \frac{1}{4}\left[0.6495 + 0 + 2(0.6495)\right] = 0.4871$$

(3) Solve $\frac{\partial^2 u}{\partial x^2} = \frac{\partial u}{\partial t}$ with $u(0,t) = u(1,t) = 0$ & $u(x,0) = x(1-x)$. Assume $h = 0.1$ tabulate u for $t = k, 2k, \& 3k$ choosing an appropriate value of k.

Soln: Compare $\frac{\partial^2 u}{\partial x^2} = \frac{\partial u}{\partial t}$ with $\frac{\partial u}{\partial t} = c^2 \frac{\partial^2 u}{\partial x^2}$

$$c^2 = 1, \quad h = 0.1, \quad \alpha = \frac{kc^2}{h^2} \Rightarrow \frac{1}{2} = \frac{k(1)}{h^2}$$

$$\Rightarrow k = \frac{h^2}{2} = \frac{(0.1)^2}{2} \quad \boxed{k = \frac{1}{200}}$$

Boundary Condition: $u(0,t) = u(1,t) = 0$.

Initial Condition: $u(x,0) = x(1-x)$

t \ x	0	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1
		0	1	2	3	4	5	6	7	8	9
0	0	u_{00}									$u_{10}=0$
$\frac{1}{200}$	1	$u_{0,\frac{1}{200}}$									$u_{1,\frac{1}{200}}$
$\frac{2}{200}$	2	$u_{0,\frac{2}{200}}$									$u_{1,\frac{2}{200}}$
$\frac{3}{200}$	3	$u_{0,\frac{3}{200}}$									$u_{1,\frac{3}{200}}$

(4) Given $\frac{\partial^2 f}{\partial x^2} - \frac{\partial f}{\partial t} = 0$; $f(0,t) = f(5,t) = 0$

$f(x,0) = x^2(25 - x^2)$ find the value of f

for $x = ih$, $i : 0, 1, 2 \ldots 5$, $t = jk$; $k = 0, 1 \ldots 6$

with $h = 1$ & $k = \frac{1}{2}$ using explicit method.

Soln: Compare $\frac{\partial^2 f}{\partial x^2} - \frac{\partial f}{\partial t} = 0$ i.e $\frac{\partial^2 f}{\partial x^2} = \frac{\partial f}{\partial t}$

with $c^2 \frac{\partial^2 u}{\partial x^2} = \frac{\partial u}{\partial t}$

$c^2 = 1$, $h = 1$, $k = 1/2$ $\quad \alpha = \frac{kc^2}{h^2} = \frac{\frac{1}{2}(1)}{1}$

$\Rightarrow \boxed{\alpha = \frac{1}{2}}$

use Bendre - Schmidt method.

$f(0,t) = f(5,t) = 0 \longrightarrow$ Boundary condition

$f(x,0) = x^2(25 - x^2) \longrightarrow$ Initial condition.

t \ x	0	1	2	3	4	5
0	$u_{00} = 0$	$u_{10} = 24$	$u_{20} = 84$	$u_{30} = 144$	$u_{40} = 144$	$u_{50} = 0$
$\frac{1}{2}$	$u_{0,\frac{1}{2}} = 0$	$u_{1,\frac{1}{2}} = 42$	$u_{2,\frac{1}{2}} = 84$	$u_{3,\frac{1}{2}} = 114$	$u_{4,\frac{1}{2}} = 72$	$u_{5,\frac{1}{2}} = 0$
$\frac{2}{2} = 1$	$u_{0,1} = 0$	$u_{1,1} = 42$	$u_{2,1} = 78$	$u_{3,1} = 78$	$u_{4,1} = 57$	$u_{5,1} = 0$
$\frac{3}{2}$	$u_{0,\frac{3}{2}} = 0$	$u_{1,\frac{3}{2}} = 39$	$u_{2,\frac{3}{2}} = 60$	$u_{3,\frac{3}{2}} = 67.5$	$u_{4,\frac{3}{2}} = 39$	$u_{5,\frac{3}{2}} = 0$
$\frac{4}{2} = 2$	$u_{0,2} = 0$	$u_{1,2} = 30$	$u_{2,2} = 53.25$	$u_{3,2} = 49.5$	$u_{4,2} = 33.75$	$u_{5,2} = 0$
$5/2$	$u_{0,\frac{5}{2}} = 0$	$u_{1,\frac{5}{2}} = 26.6$	$u_{2,\frac{5}{2}} = 39.75$	$u_{3,\frac{5}{2}} = 43.5$	$u_{4,\frac{5}{2}} = 24.75$	$u_{5,\frac{5}{2}} = 0$
$\frac{6}{2} = 3$	$u_{03} = 0$	$u_{1,3} = 19.88$	$u_{2,3} = 35.06$	$u_{3,3} = 32.25$	$u_{4,3} = 21.75$	$u_{5,3} = 0$

$u(x,0) = x^2(25-x^2)$

$x=1 \quad u(1,0) = 24$

$x=2 \quad u(2,0) = 84$

$x=3 \quad u(3,0) = 144$

$x=4 \quad u(4,0) = 144$

II row

$$u_{1,\frac{1}{2}} = \frac{1}{2}\left[u_{00} + u_{2,0}\right] = 42$$

$$u_{2,\frac{1}{2}} = \frac{1}{2}\left[u_{1,0} + u_{3,0}\right] = 84$$

$$u_{3,\frac{1}{2}} = \frac{1}{2}\left[u_{2,0} + u_{4,0}\right] = 114$$

$$u_{4,\frac{1}{2}} = \frac{1}{2}\left[u_{3,0} + u_{5,0}\right] = 72.$$

III ly for 3rd, 4th, 5th, 6th & 7th rows.

[go up add [left & right] ÷ 2]

HW

① Solve the heat eqn $\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2}$ subject to the condition $u(0,t) = u(1,t) = 0$ & $u(x,0) = \begin{cases} 2x & 0 \le x < \frac{1}{2} \\ 2(1-x) & \frac{1}{2} \le x \le 1 \end{cases}$

taking $h = \frac{1}{4}$ & k according to Bendre-Schmidt eqn $T = 3$ (level in t).

Soln: Since Bendre Schmidt $\alpha = \frac{1}{2}$

compare $c^2 = 1$, $h = \frac{1}{4}$ $\alpha = \frac{kc^2}{h^2}$, $\frac{1}{2} = \frac{k(1)}{\frac{1}{16}}$ $16k = \frac{1}{2}$ $k = \frac{1}{32}$

Since $T = 3$ $0 - \frac{1}{32} \to \frac{2}{32} \to \frac{3}{32}$ (3 levels)

483338
OTP · 3581

Numerical soln of wave eqn [Hyperbolic eqn]

The standard equation is of the form

$$\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2} \longrightarrow (1)$$

phase speed $c^2 = \frac{T}{\rho}$ Tension. Ratio of tension to mass/unit length of string. ρ: density of the string.

Subject to initial conditions: $u = f(x)$, $\frac{\partial u}{\partial t} = g(x)$

$0 \le x \le l$ at $t = 0 \Rightarrow u(x,0) = f(x)$ & $u_t(x,0) = g(x)$

& the boundary conditions

$u(0,t) = \phi(t)$, $u(l,t) = \psi(t)$.

Consider a rectangular mesh in the x-t plane spacing h along x direction & k along time t direction. Denoting a mesh point $(x,t) = i,j$ as simply.

Substitute $\frac{\partial^2 u}{\partial x^2} = \frac{u_{i-1,j} - 2u_{i,j} + u_{i+1,j}}{h^2}$ &

$\frac{\partial^2 u}{\partial t^2} = \frac{u_{i,j-1} - 2u_{i,j} + u_{i,j+1}}{k^2}$ using

finite difference approximation in the standard eqn (1). After simplification it reduces to $u_{i,j+1} = 2(1-\alpha^2c^2)u_{i,j} + \alpha^2c^2(u_{i-1,j} + u_{i+1,j}) - u_{i,j-1} \longrightarrow (2)$ where $\alpha = \frac{k}{h}$.

The coefficient of $u_{i,j}$ vanishes if $\alpha c = 1$ then (2) $\Rightarrow$